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Structured Debt Ratings: Evidence on Conflicts of Interest

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1 Big picture

- **Question.** Do credit rating agencies give more favorable ratings to issuers that provide them with more securitization business?
- **Setting.** The paper studies structured debt securities - ABS, MBS, CDOs, and CLOs - issued between 1999 and 2011 and rated by Moody's, S&P, or Fitch.
- **Conflict-of-interest channel.** Rating agencies are paid by issuers. When a CRA receives a large volume of business from a given issuer, the issuer becomes a key client and the CRA has a stronger incentive to preserve the relationship by granting rating favors.
- **Main challenge.** A favorable rating could reflect better collateral quality, stronger credit enhancement, better issuer reputation, or true lower credit risk. The paper therefore controls for deal-level credit risk and issuer fixed effects.
- **Key innovation.** The authors construct a *deal-level rating* statistic that summarizes the full distribution of tranche ratings below the AAA cut-off. The statistic is built from rating-implied spreads, so it has a cardinal interpretation in basis-point-like units.
- **Main result.** Conditional on deal characteristics, credit enhancement, asset type, collateral nationality, time effects, and issuer fixed effects, larger shared business between an issuer and a CRA is associated with lower deal rating spreads, i.e., better ratings.
- **Economic magnitude.** In the preferred specification, a one-standard-deviation increase in log shared business reduces the deal rating by about 41%, corresponding to roughly a four basis point improvement for a representative deal.
- **Complexity result.** Rating favors are stronger for more complex structured products, proxied by a higher number of tranches.
- **Credit-cycle result.** Rating favors are larger during the 2004–2006 credit boom and less pronounced during the 2007–2008 crisis.
- **Agency heterogeneity.** All three major CRAs show the conflict pattern. The relationship-based favors appear more pronounced for S&P and Fitch than for Moody's.
- **Robustness.** The paper also studies AAA subordination and nonparametric comparisons of top-client deals. Large clients receive lower AAA subordination and better deal ratings, even though their collateral quality is not better.
- **Takeaway.** Structured-debt ratings were influenced by issuer-CRA business relationships. The evidence supports the view that issuer-pay ratings can generate systematic rating favors, especially when products are opaque and boom-time reputational costs are low.

2 Data and measurement

2.1 Structured debt sample

- The paper uses structured debt securities backed by collateral from North America, Europe, the Middle East, or Africa.
- The security-level sample contains 22,359 securitized tranches from 7,255 deals, with total issuance volume of about USD 6,968 billion.
- The securities are issued by 949 issuers and are reported by DCM Analytics.
- The authors observe launch ratings by Moody's, S&P, and Fitch. When a security has multiple ratings, the more conservative rating is used for the composite rating.
- The deal-level regression sample has 672 deals and 1,404 deal-CRA observations with full data coverage.

Interpretation. The analysis is at the deal-CRA level, not the individual tranche level, because many credit-risk controls are naturally measured for the full collateral pool or deal.

2.2 Market composition

- Residential MBS account for around 60% of issuance volume.
- Commercial MBS account for around 5%.
- CDOs and CLOs account for roughly 3% and 5%.
- By collateral nationality, the United States is the largest group, followed by the United Kingdom and Spain.
- Issuance grows sharply after 2004, peaks near 2007, and falls after the financial crisis begins.

Interpretation. The sample covers the securitization boom-bust cycle, which is central for testing whether rating favors are stronger in good times.

2.3 Tranche ratings and rating-implied spreads

The authors convert ordinal ratings into a cardinal rating statistic using a first-stage spread regression:

$$y = \alpha_D D + \alpha_Z Z_{tr} + \epsilon, \quad (1)$$

where y is the yield spread at issuance, D is a vector of rating dummies, and Z_{tr} contains tranche, deal, and market controls.

For a given rating R , the rating-implied spread is

$$RIS_R = (\hat{\alpha}_D)_R. \quad (2)$$

Interpretation. A rating-implied spread translates a rating category into the average spread associated with that rating. Better ratings imply lower RIS values.

2.4 Deal rating

For deal d , CRA a , and tranche tr , let ω_{tr} be the tranche's face-value weight. The deal-level rating statistic is

$$Deal\ rating(d, a) = \sum_{tr=1}^n \omega_{tr} RIS(a, d, tr). \quad (3)$$

- Lower values mean better ratings.

- The statistic uses the full distribution of rated and unrated tranches.
- Because units are yield-spread units, changes can be interpreted in basis-point-like terms.

Interpretation. The statistic avoids a simple numerical average of ratings, which would be inappropriate because rating categories are ordinal and nonlinear.

2.5 Shared business

The key relationship variable is the securitization volume that issuer d shares with CRA a in a 12-month window around issuance:

$$Shared\ business(d, a) = \sum_{tr \in \Omega(d, a)} face\ value(tr), \quad (4)$$

where $\Omega(d, a)$ is the set of all tranches rated by CRA a and sold by issuer d in the months $[t - 6, t + 6]$ around the deal.

Interpretation. Shared business is bilateral. The same issuer can be a key client of one CRA and a less important client of another CRA. This is the core source of identifying variation.

2.6 Credit-risk and deal controls

The baseline controls include:

- collateral delinquency rate and its squared term;
- deal fraction with guarantee;
- overcollateralization;
- reserve fund;
- number of tranches;
- deal face value;
- time fixed effects;
- collateral nationality fixed effects;
- asset-type fixed effects;
- issuer fixed effects.

Interpretation. These controls are designed to separate rating favors from true differences in deal risk and issuer quality.

3 Models and empirical specifications

3.1 Baseline relationship-ties regression

The main regression is

$$\log Deal\ rating(d, a) = \beta_{SB} \log Shared\ business(d, a) + \beta_C C(d) + \varepsilon_{d,a}, \quad (5)$$

where $C(d)$ contains deal controls and fixed effects.

- d indexes deals.
- a indexes credit rating agencies.
- The dependent variable is the log deal-level rating statistic.
- A lower deal rating means a better rating.
- Standard errors are clustered both by deal and by issuer.

Interpretation. The coefficient of interest is β_{SB} . A negative coefficient means stronger issuer-CRA business ties are associated with better ratings for the same issuer's deals.

3.2 Complexity interaction

To test whether rating favors are stronger when products are harder to evaluate, the paper estimates

$$\log Deal\ rating(d, a) = \beta_1 \log Shared\ business(d, a) + \beta_2 (No.\ of\ tranches_d \times \log Shared\ business(d, a)) + \Gamma C(d) + \varepsilon_{d,a}. \quad (6)$$

Interpretation. A negative β_2 means the effect of shared business is stronger for more complex deals.

3.3 Credit-cycle interactions

The paper interacts shared business with boom and crisis-period dummies:

$$\begin{aligned} \log Deal\ rating(d, a) = & \beta_1 \log Shared\ business(d, a) \\ & + \beta_2 (Issued\ 2004-2006_d \times \log Shared\ business(d, a)) \\ & + \beta_3 (Issued\ 2007-2008_d \times \log Shared\ business(d, a)) + \Gamma C(d) + \varepsilon_{d,a}. \end{aligned} \quad (7)$$

Interpretation. If rating favors are larger during booms, β_2 should be negative. If reputational costs rise during crises, β_3 should be weaker or insignificant.

3.4 Agency heterogeneity

To test whether the effect differs across CRAs, the paper adds CRA dummies and interactions:

$$\log Deal\ rating(d, a) = \beta \log Shared\ business(d, a) + \beta_S (S\&P_a \times \log Shared\ business(d, a)) + \beta_F (Fitch_a \times \log Shared\ business(d, a)) + \Gamma C(d) + \varepsilon_{d,a}. \quad (8)$$

Interpretation. Negative interaction coefficients for S&P and Fitch imply larger relationship-based rating favors relative to Moody's.

3.5 AAA subordination robustness

The paper also replaces the deal rating with the AAA subordination level:

$$AAA\ subordination(d, a) = \frac{\text{deal face value below AAA tranches}}{\text{total deal face value}}. \quad (9)$$

Interpretation. Lower AAA subordination means that a larger fraction of the deal receives AAA treatment. If top-client issuers receive lower AAA subordination, the result is consistent with rating favors.

4 Identification and empirical design

4.1 What identifies the effect

- Identification comes from variation across CRA-issuer pairs in the amount of bilateral securitization business.
- Issuer fixed effects absorb issuer reputation, securitization expertise, and other issuer-level characteristics.
- Since the same issuer can have different business ties with different CRAs, the paper can compare ratings for the same issuer across agencies.
- Deal-level controls absorb observable credit-risk differences.

Interpretation. The strongest design feature is that unobserved issuer quality does not mechanically explain the findings once issuer fixed effects are included.

4.2 Why the deal-level rating statistic matters

- Rating favors in structured products may appear not only in the headline AAA label, but also in the distribution of tranche ratings below AAA.
- A deal-level statistic uses all tranche ratings rather than focusing on only one tranche.
- Using rating-implied spreads gives the statistic a cardinal interpretation.

Interpretation. The statistic is designed to capture subtle rating inflation across the entire deal structure.

4.3 Main empirical threat: omitted credit risk

- Strong issuers may have better collateral pools or better credit enhancement.
- If so, they should receive better ratings even without conflicts of interest.
- The paper addresses this by controlling for delinquency, guarantees, overcollateralization, reserve funds, number of tranches, asset type, collateral nationality, and time.

Interpretation. The credibility of the result depends on whether these controls capture the main sources of deal credit risk.

4.4 Why complexity and cycle tests are informative

- Rating favors should be easier to hide in more complex deals.
- They should also be more tempting in boom periods when default probabilities and reputational costs seem lower.
- The paper finds stronger effects in both settings.

Interpretation. These heterogeneity results strengthen the conflict-of-interest interpretation because they line up with the theory of when rating favors should be most likely.

4.5 What the paper cannot fully prove

- The paper is observational and does not randomly assign issuer-CRA relationships.
- Some unobserved credit-risk characteristics may remain even after extensive controls.
- Shared business measures relationship strength but may also capture strategic issuer-agency matching.
- The paper focuses on launch ratings, not every subsequent downgrade or surveillance action.

Interpretation. The evidence is strongest as a carefully controlled association consistent with conflicts of interest, supported by multiple robustness and heterogeneity tests.

5 Tables and figures

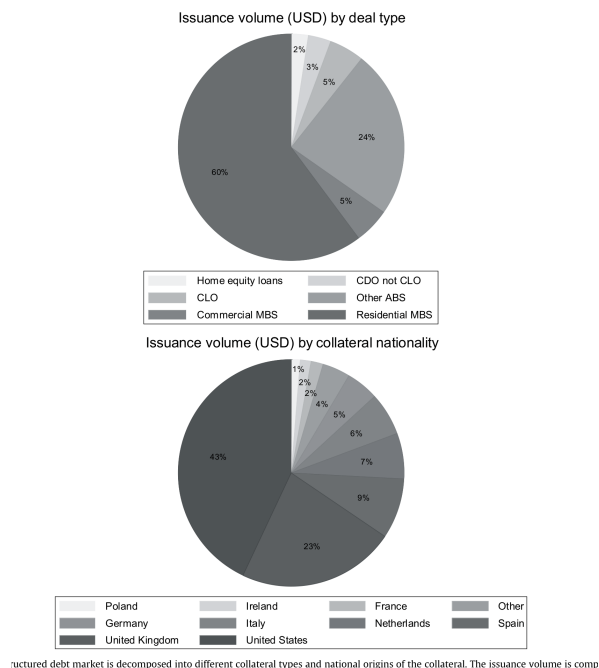
This section keeps the same *Read* plus *Economic message* format. Adjacent figures and tables are grouped to save space and improve comparison. All visuals are cropped directly from the original paper.

5.1 Figures 1–2: market composition and growth

Read:

- Figure 1 decomposes issuance by deal type and collateral nationality.
- Residential MBS represent the largest deal type by issuance volume.
- US collateral is the largest national group, followed by the UK and Spain.
- Figure 2 shows the boom-bust path of structured-debt issuance from 1999 to 2011.
- Issuance rises sharply after 2004, peaks in 2007, and contracts after the crisis begins.

Economic message. The paper studies a market that grew rapidly during the credit boom and then collapsed. This timing is important because conflicts of interest should be most visible when the market is hot and products are opaque.



structured debt market is decomposed into different collateral types and national origins of the collateral. The issuance volume is comp

(a) Figure 1: issuance by deal type and collateral nationality.

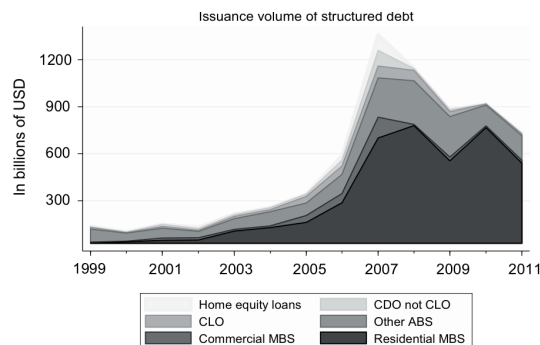


Fig. 2. The market growth of structured debt between 1999 and 2011. The issuance volume is computed as the total principal value in USD of all securities in a given year. The sample comprises all securities that have collateral from North America, Europe, Middle East, or Africa, are issued between 1999 and 2011, have an ISIN identifier, and are reported by DCM Analytics. The total issuance volume is USD 6,968bn.

(b) Figure 2: structured-debt issuance volume over time.

5.2 Tables 1–2: tranche-level and deal-level summary statistics

Read:

- Table 1 shows that 46% of tranches receive AAA composite ratings, 11% receive AA ratings, 13% receive A ratings, 12% receive BBB ratings, and 5% are junk.
- The average yield spread is 89 basis points and average tranche face value is USD 312 million.
- Table 2 reports the deal-level sample: 672 deals and 1,404 deal-CRA observations; average deal face value is USD 2.08 billion, and average shared business is USD 10.20 billion.

Economic message. The sample has substantial variation in ratings, credit risk, credit enhancement, and issuer-CRA business relationships.

Table 1

Tranche-level summary statistics.

Reported are summary statistics on the tranche-level characteristics, the market term structure data at issuance, and the imputed *Rating-implied spread* (*RIS*). Composite ratings are determined for securities with ratings from more than one CRA: If a security has two ratings, the more conservative rating is documented. If the security has three ratings, the median rating is chosen. PDS is short for Moody's database Performance Data Service, DCM is short for DCM Analytics, Bloomb. stands for Bloomberg, Datastr. stands for Thomson Reuters Datastream, and calcul. abbreviates calculated.

Variable	Description	Source	Obs	Mean	Median	Std. dev.	Min	Max
<i>Panel A: Tranche characteristics</i>								
<i>Yield spread</i>	To Euribor or Libor in %	DCM	10,625	0.89	0.50	1.04	0.03	4.50
<i>Tranche face value</i>	Face value in USD	DCM	22,359	312m	71m	867m	404	64.7bn
<i>Log tranche face value</i>		calcul.	22,359	18.20	18.08	1.76	6.00	24.89
<i>Issuance price</i>	In % of face value	Bloomb.	13,427	99.81	100	3.45	1.09	214.85
<i>Weighted avg. life</i>	At issuance in years	Bloomb.	17,706	5.55	4.97	3.62	0.1	33.82
<i>Panel B: Composite rating dummies</i>								
<i>AAA</i>	1 for ratings shown	calcul.	22,359	0.46	–	–	–	–
<i>AA+ /AA/AA–</i>	1 for ratings shown	calcul.	22,359	0.11	–	–	–	–
<i>A+ /A/A–</i>	1 for ratings shown	calcul.	22,359	0.13	–	–	–	–
<i>BBB+ /BBB/BBB–</i>	1 for ratings shown	calcul.	22,359	0.12	–	–	–	–
<i>Junk</i>	Rating below BBB–	calcul.	22,359	0.05	–	–	–	–
<i>Unrated senior/mezzanine/junior</i>	Unrated, not subordinated to any <i>rated</i> tranche	calcul.	22,359	0.09	–	–	–	–
<i>Unrated mezzanine/junior</i>	Unrated, only subordinated to <i>rated</i> mezzanine tranche	calcul.	22,359	0.02	–	–	–	–
<i>Unrated junior</i>	Unrated, subordinated to <i>rated</i> junior tranche	calcul.	22,359	0.01	–	–	–	–
<i>Panel C: Term structure at issuance (in %)</i>								
<i>Term structure level</i>	1mth US Libor	Datastr.	22,355	3.27	3.58	2.09	0.19	6.82
<i>Term structure slope</i>	12mth minus 1mth US Libor	Datastr.	22,355	0.34	0.28	0.44	–0.82	1.73
<i>Panel D: Rating implied credit spread (in %)</i>								
<i>RIS(Moody's)</i>	Implied by Moody's rating	calcul.	22,359	0.35	0.26	0.53	0.00	2.61
<i>RIS(S&P)</i>	Implied by S&P rating	calcul.	22,359	0.35	0.06	0.57	0.00	2.61
<i>RIS(Fitch)</i>	Implied by Fitch rating	calcul.	22,359	0.36	0.44	0.41	0.00	2.61

(a) Table 1: tranche-level summary statistics.

Table 2

Deal-level summary statistics.

Reported are summary statistics on the deal characteristics, the imputed *Deal rating*, AAA subordination levels, and *Shared business* for the final regression sample. *Delinquency* is measured nine months after deal closure. If no observation with nine months' seasoning exists, the delinquency observation closest to nine months' seasoning (at least six and at most 12 months' seasoning accepted) is chosen and linearly adjusted (see Internet Appendix). For *Overcollateralization* and *Reserve fund* the youngest available observation after deal closure is chosen (at most six months' seasoning accepted). PDS is short for Moody's database Performance Data Service, DCM is short for DCM Analytics, and calcul. abbreviates calculated.

Variable	Description	Source	Obs	Mean	Median	Std. dev.	Min	Max
<i>Panel A: Deal characteristics</i>								
<i>Deal face value</i>	Face value in USD	DCM	672	2.08bn	1.18bn	4.01bn	7710m	69.5bn
<i>Log deal face value</i>	Natural log of deal face value	calcul.	672	20.94	20.89	0.91	18.16	24.96
<i>Delinquency</i>	Winsorized fraction (in %) of delinquent collateral 9mth after deal closure	PDS	672	1.29	0.35	2.37	0.00	12.17
<i>Log delinquency</i>	Log(Del. + 0.0053)	calcul.	672	–1.01	–1.04	1.70	–5.23	2.50
<i>Deal fraction with guarantee</i>	Winsorized ratio of guaranteed principal over deal face value	DCM	672	0.06	0.00	0.21	0.00	1.00
<i>Overcollateral.</i>	Winsorized ratio of collateral principal minus principal of securities over principal of securities	PDS	672	–0.003	0	0.05	–0.13	0.21
<i>Reserve fund</i>	Winsorized reserves divided by principal of securities	PDS	672	0.03	0.02	0.03	0	0.13
<i>Number of tranches</i>	No. of deal tranches	DCM	672	4.84	4	2.90	1	26
<i>Unsecuritized deal part</i>	Ratio of deal face value minus principal of securities over deal face value	calcul.	672	0.01	0	0.07	0.00	0.98
<i>Single CRA</i>	1 if all tranche ratings from same CRA	calcul.	672	0.24	–	–	–	–
<i>Panel B: Deal rating (in %)</i>								
<i>Deal rating (S&P)</i>	Implied by S&P ratings	calcul.	433	0.10	0.04	0.21	0	1.98
<i>Deal rating (Moody's)</i>	Implied by Moody's ratings	calcul.	620	0.11	0.04	0.20	0	1.98
<i>Deal rating (Fitch)</i>	Implied by Fitch ratings	calcul.	351	0.10	0.04	0.21	0	1.98
<i>Deal rating</i>	<i>Deal rating</i> of all deal-CRA pairs	calcul.	1,404	0.10	0.04	0.21	0	1.98
<i>Log deal rating</i>	Log(<i>Deal rating</i> + 0.0030)	calcul.	1,404	–3.14	–3.17	1.29	–5.82	0.68
<i>Panel C: AAA subordination levels</i>								
<i>AAA subord. (S&P)</i>	Deal fraction not AAA by S&P	calcul.	433	0.22	0.08	0.32	0	1
<i>AAA subord. (Moody's)</i>	Deal fraction not Aaa by Moody's	calcul.	620	0.19	0.08	0.29	0	1
<i>AAA subord. (Fitch)</i>	Deal fraction not AAA by Fitch	calcul.	351	0.13	0.08	0.16	0	1
<i>AAA subord.</i>	Deal fraction not AAA by any CRA	calcul.	1,404	0.19	0.08	0.28	0	1
<i>Log AAA subord.</i>	Ln(<i>AAA subord.</i> + 0.0070)	calcul.	1,404	–2.35	–2.44	1.17	–4.96	0.01
<i>Panel D: Shared Business</i>								
<i>Shared business</i>	Business (USD) between CRA <i>a</i> and issuer of deal <i>d</i> over 12 months	calcul.	1,404	10.20bn	3.60bn	18.2bn	12.6m	120bn
<i>Log shared business</i>	Natural log of <i>Shared business</i>	calcul.	1,404	22.04	22.00	1.44	16.35	25.51

(b) Table 2: deal-level summary statistics.

5.3 Tables 3–4: rating-implied spreads and issuer-CRA relationship heterogeneity

Read:

- Table 3 estimates how tranche ratings map into issuance yield spreads. The relationship is nonlinear; junk and unrated junior tranches imply much larger spreads.
- Column 4 includes tranche characteristics, maturity, term-structure variables, and fixed effects, with $R^2 = 0.733$.
- Table 4 shows that top issuers differ across CRAs and that the same issuer can generate very different business volumes for different agencies.

Economic message. Table 3 constructs the rating-implied spread input for the deal rating. Table 4 shows the bilateral variation that makes issuer fixed effects feasible.

Table 3
Estimating rating-implied spreads.
We regress tranche-level yield spread on tranche rating and various control variables, which include *Log tranche face value* as well as its squared value; the *Weighted average life* of the tranche at issuance as well as its squared value; the one-month USD Libor rate at issuance as a proxy for the *Term structure level*; the difference between the 12-month and the one-month USD Libor rate at tranche issuance as a proxy for the *Term structure slope*. Time fixed, asset-type fixed, collateral nationality fixed, and currency fixed effects as well as their time-interactions are included in columns 2 to 4. Standard errors (in parentheses) are clustered for deals. The symbols *, **, and *** represent significance levels of 10%, 5%, and 1%, respectively.

	Dependent variable: Yield spread			
	(1)	(2)	(3)	(4)
Rating dummies:				
AA+	0.114*** (0.040)	0.090*** (0.032)	−0.005 (0.038)	−0.002 (0.038)
AA	0.148*** (0.018)	0.004 (0.013)	0.182*** (0.019)	0.008 (0.019)
AA−	0.196*** (0.036)	0.248*** (0.027)	0.059* (0.031)	0.059* (0.031)
A+	0.434*** (0.055)	0.463*** (0.037)	0.245*** (0.039)	0.241*** (0.038)
A	0.407*** (0.021)	0.472*** (0.016)	0.260*** (0.023)	0.264*** (0.023)
A−	0.627*** (0.060)	0.587*** (0.047)	0.280*** (0.049)	0.282*** (0.050)
BBB+	1.014*** (0.104)	1.091*** (0.078)	0.720*** (0.076)	0.718*** (0.075)
BBB	0.961*** (0.034)	1.056*** (0.028)	0.776*** (0.033)	0.780*** (0.033)
BBB−	1.060*** (0.057)	1.195*** (0.044)	0.883*** (0.048)	0.885*** (0.048)
Junk	2.818*** (0.047)	2.933*** (0.043)	2.607*** (0.052)	2.609*** (0.052)
Unrated senior/mezzanine/junior	0.666*** (0.047)	0.598*** (0.052)	0.434*** (0.056)	0.441*** (0.055)
Unrated mezzanine/junior	1.457*** (0.189)	1.089*** (0.153)	0.995*** (0.179)	0.999*** (0.178)
Unrated junior	2.443*** (0.316)	2.189*** (0.318)	2.013*** (0.379)	2.016*** (0.378)
Controls:				
Log tranche face value			−0.617*** (0.126)	−0.617*** (0.127)
Log tranche face value squared			0.014*** (0.003)	0.014*** (0.003)
Weighted avg. life			0.025*** (0.006)	0.024*** (0.006)
Weighted avg. life squared			−0.001*** (0.000)	−0.001*** (0.000)
Term structure level			−0.159*** (0.028)	−0.159*** (0.028)
Term structure slope			−0.241*** (0.043)	−0.241*** (0.043)
Fixed effects & interactions:	No	Yes	Yes	Yes
R ²	0.484	0.703	0.730	0.733
N	10,675	10,675	9,314	9,314

(a) Table 3: estimating rating-implied spreads.

Table 4
Top 10 issuers by credit rating agency.
Panel A shows the top 10 issuers according to the total securitization business shared between a given issuer and CRA over the years 1999 to 2011. Panel B shows total *Shared business* (aggregated over the years 1999 to 2011) in USD bn for each credit rating agency and issuer listed in Panel A.

Panel A: Top 10 issuers ranked by total Shared business (1999 to 2011)			
Rank	Moody's	S&P	Fitch
1	Banco Santander	Banco Santander	Lloyds Banking
2	Lloyds Banking	Lloyds Banking	Banco Santander
3	Royal Bank of Scotland	Royal Bank of Scotland	Royal Bank of Scotland
4	Barclays	Bank of America	Northern Rock
5	Bank of America	Barclays	Barclays
6	Northern Rock	JPMorgan	JPMorga
7	Rabobank Nederland	Northern Rock	Bank of America
8	JPMorgan	Lehman Brothers	ABN AMRO Bank
9	Lehman Brothers	ABN AMRO Bank	Lehman Brothers
10	ABN AMRO Bank	Deutsche Bank	Nationwide Building Society
Panel B: Total Shared business (1999 to 2011)			
Issuer	Moody's	S&P	Fitch
Banco Santander	285	314	248
Lloyds Banking	264	286	269
Royal Bank of Scotland	191	164	164
Barclays	164	125	97
Bank of America	136	157	89
Northern Rock	109	109	101
Rabobank Nederland	105	32	21
JPMorgan	102	110	89
Lehman Brothers	88	102	67
ABN AMRO Bank	84	84	69
Nationwide Building Society	52	52	52
Deutsche Bank	25	67	22

(b) Table 4: top issuers by credit rating agency.

5.4 Tables 5–6: main rating-favor evidence, complexity, cycles, and agency effects

Read:

- Table 5 is the main regression. The coefficient on log shared business is negative in all specifications; with issuer fixed effects, column 2 gives -0.285 , significant at the 1% level.
- Column 4 keeps both log shared business and log deal face value; shared business remains important while deal size becomes insignificant.
- Table 6 shows stronger favors for complex deals: number of tranches $\times$ log shared business is -0.071 and significant. The 2004–2006 boom interaction is -0.295 and significant.
- S&P and Fitch interactions with log shared business are negative, suggesting larger relationship-based favors relative to Moody's.

Economic message. The same issuer receives better ratings from the CRA with which it shares more business, especially for complex boom-period deals.

Table 5

Relationship ties and rating favors.

The (log) *Deal rating* is regressed on the (log) *Shared business*, which proxies the strength of the relationship ties between a given CRA and the issuer of a deal. The controls are: *Log deal face value* = natural logarithm of deal face value; *Log delinquency* = logarithm of collateral delinquency as well as its squared value; *Deal fraction with guarantee* = face value of guaranteed tranches divided by deal face value; *Overcollateralization* = difference between collateral and securities' principal divided by principal of securities; *Reserve fund* = liquidity reserves standardized by principal of securities; *No. of tranches* = number of deal tranches; constant (unreported). Column 5 uses the full sample of deals (except deals issued by Fannie Mae, Freddie Mac, or Ginnie Mae). If data on the delinquency rate, overcollateralization, or reserve funds are missing, we set these values to zero (only in column 5) and include dummies that indicate missing values as additional controls. Standard errors (in parentheses) are clustered in two dimensions both by deal and by issuer. The symbols *, **, and *** represent significance levels of 10%, 5%, and 1%, respectively.

	Dependent variable: <i>Log deal rating</i>				
	(1)	(2)	(3)	(4)	(5)
<i>Log shared business</i>	-0.195** (0.078)	-0.285*** (0.083)		-0.273*** (0.081)	-0.234*** (0.049)
<i>Log deal face value</i>			-0.158* (0.088)	-0.070 (0.085)	-0.169** (0.072)
Controls:					
<i>Log delinquency</i>	0.216*** (0.081)	0.234*** (0.082)	0.203*** (0.077)	0.228*** (0.081)	0.035 (0.051)
<i>Log delinquency squared</i>	0.036** (0.018)	0.044** (0.017)	0.040** (0.017)	0.042** (0.017)	0.001 (0.013)
<i>Deal fraction with guarantee</i>	-0.531** (0.244)	-0.310* (0.177)	-0.301 (0.201)	-0.342* (0.177)	-0.489*** (0.162)
<i>Overcollateralization</i>	-1.248 (1.105)	-0.384 (1.515)	-0.333 (1.509)	-0.392 (1.506)	0.323 (1.112)
<i>Reserve fund</i>	3.236 (3.870)	1.097 (3.815)	2.279 (3.816)	1.210 (3.777)	1.707 (3.433)
<i>No. of tranches</i>	0.151*** (0.038)	0.142*** (0.033)	0.134*** (0.036)	0.146*** (0.035)	0.205*** (0.020)
Time fixed effects	Yes	Yes	Yes	Yes	Yes
Collateral nationality fixed effects	Yes	Yes	Yes	Yes	Yes
Asset-type fixed effects	Yes	Yes	Yes	Yes	Yes
Issuer fixed effects	No	Yes	Yes	Yes	Yes
Dummies for missing values					Yes
R^2	0.316	0.551	0.536	0.551	0.535
N	1,404	1,404	1,404	1,404	8,544

(a) Table 5: relationship ties and rating favors.

Table 6

Asset complexity, credit cycles, and agency fixed effects.

The (log) *Deal rating* is regressed on the (log) *Shared business*, which proxies the strength of the relationship ties between a given CRA and the issuer of a deal. Column 1 includes an interaction term between *Log shared business* and the *No. of tranches* in the deal. Column 2 includes two interaction terms between credit boom and crisis dummies and *Log shared business*. Column 3 includes CRA fixed effects and column 4 additionally includes interactions between CRA dummies and *Log shared business*. The controls are: *Log delinquency* = logarithm of collateral delinquency as well as its squared value; *Deal fraction with guarantee* = face value of guaranteed tranches divided by deal face value; *Overcollateralization* = difference between collateral and securities' principal divided by principal of securities; *Reserve fund* = liquidity reserves standardized by principal of securities; *No. of tranches* = number of deal tranches; constant (unreported). Standard errors (in parentheses) are clustered in two dimensions both by deal and by issuer. The symbols *, **, and *** represent significance levels of 10%, 5%, and 1%, respectively.

	Dependent Variable: <i>Log deal rating</i>			
	(1)	(2)	(3)	(4)
<i>Log shared business</i>	0.070 (0.103)	-0.173** (0.084)	-0.289*** (0.084)	-0.245*** (0.082)
<i>No. of tranches $\times$ Log shared business</i>	-0.071*** (0.018)			
<i>Issued 2004–2006 $\times$ Log shared business</i>		-0.295** (0.120)		
<i>Issued 2007–2008 $\times$ Log shared business</i>		-0.074 (0.108)		
Dummy: S&P			-0.091 (0.055)	1.543** (0.671)
Dummy: Fitch			-0.036 (0.041)	1.807*** (0.682)
<i>S&P $\times$ Log shared business</i>				-0.074*** (0.030)
<i>Fitch $\times$ Log shared business</i>				-0.083*** (0.031)
Controls:				
<i>Log delinquency</i>	0.203** (0.080)	0.237*** (0.079)	0.235*** (0.083)	0.224*** (0.082)
<i>Log delinquency squared</i>	0.044*** (0.016)	0.043*** (0.017)	0.044*** (0.017)	0.044*** (0.017)
<i>Deal fraction with guarantee</i>	-0.398*** (0.165)	-0.299 (0.182)	-0.312* (0.178)	-0.317* (0.179)
<i>Overcollateralization</i>	-0.149 (1.419)	-0.294 (1.496)	-0.372 (1.511)	-0.343 (1.504)
<i>Reserve fund</i>	2.878 (3.348)	0.457 (3.932)	1.082 (3.815)	1.109 (3.817)
<i>No. of tranches</i>	1.808*** (0.425)	0.144*** (0.033)	0.142*** (0.033)	0.143*** (0.033)
Time fixed effects	Yes	Yes	Yes	Yes
Collateral nationality fixed effects	Yes	Yes	Yes	Yes
Asset-type fixed effects	Yes	Yes	Yes	Yes
Issuer fixed effects	Yes	Yes	Yes	Yes
R^2	0.577	0.559	0.551	0.553
N	1,404	1,404	1,404	1,404

(b) Table 6: complexity, credit cycles, and CRA fixed effects.

5.5 Tables 7–8: asset types and nonparametric robustness

Read:

- Table 7 shows that the relationship-ties effect is present for both MBS and ABS, though stronger for MBS; the ABS interaction in the full sample is not statistically significant.
- Table 8 compares top CRA clients against the rest without imposing the baseline regression structure. Top clients receive lower AAA subordination and lower deal ratings from all three CRAs.
- These top-client deals do not appear safer; delinquency rates are higher for top clients in several comparisons, and their deals are larger.

Economic message. The robustness evidence points in the same direction as the regressions: strong CRA clients receive more favorable ratings and lower AAA subordination.

Table 7

Conflicts of interest across asset types.

The (log) *Deal rating* is regressed on the (log) *Shared business*, which proxies the strength of the relationship ties between a given CRA and the issuer of deal. Column 1 considers the subsample of MBS and column 2 the subsample of ABS. Column 3 considers the full sample and includes an interaction of *Lc shared business* with a dummy equal to one for ABS deals. Further dummies for commercial and residential MBS as well as for CLOs are included. *Tl* controls are: *Log delinquency*=logarithm of collateral delinquency as well as its squared value; *Deal fraction with guarantee*=face value of guarantee tranches divided by deal face value; *Overcollateralization*=difference between collateral and securities' principal divided by principal of securities; *Reserve fund*=liquidity reserves standardized by principal of securities; *No. of tranches*=number of deal tranches; constant (unreported). Standard errors (i parentheses) are clustered in two dimensions both by deal and by issuer. The symbols **, and *** represent significance levels of 10%, 5%, and 1% respectively.

	Dependent Variable: <i>Log deal rating</i>		
	MBS (1)	ABS (2)	All (3)
<i>Log shared business</i>	−0.341*** (0.094)	−0.185* (0.109)	−0.341*** (0.096)
<i>ABS × Log shared business</i>			0.128 (0.134)
Controls:			
<i>Log delinquency</i>	0.215** (0.108)	0.342* (0.207)	0.215* (0.110)
<i>Log delinquency squared</i>	0.017 (0.016)	0.124*** (0.043)	0.017 (0.016)
<i>Deal fraction with guarantee</i>	−0.205 (0.206)	−1.269 (0.851)	−0.205 (0.211)
<i>Overcollateralization</i>	−1.886 (1.899)	2.580** (1.199)	−1.886 (1.938)
<i>Reserve fund</i>	−3.186 (6.039)	14.057** (5.637)	−3.186 (6.165)
<i>No. of tranches</i>	0.130*** (0.032)	0.274*** (0.074)	0.130*** (0.033)
Dummy: residential MBS			2.836 (2.938)
Dummy: commercial MBS	1.073** (0.333)		3.909 (3.001)
Dummy: CLO		−0.115 (0.499)	−0.403 (0.493)
Time fixed effects	Yes	Yes	Yes
Collateral nationality fixed effects	Yes	Yes	Yes
Issuer fixed effects	Yes	Yes	Yes
Interactions: <i>ABS dummy × Controls</i>	No	No	Yes
<i>R</i> ²	0.540	0.678	0.622
<i>N</i>	1,059	345	1,404

(a) Table 7: conflicts of interest across asset types.

Table 8

Non-parametric test for rating favors.

Reported are average deal characteristics, the number of observations, and test results by CRA for (i) all deals rated, (ii) only the deals issued by the top 10% clients, and (iii) only the deals issued by the top 5% clients of the CRA in question. The top clients of a CRA are the largest issuers as identified by the securitization volume that they asked the CRA to rate between 1999 and 2011. Deals issued by Freddie Mac, Fannie Mae, and Ginnie Mae are excluded. The values of the variables *AAA subordination*, *Deal rating*, *Delinquency rate*, *Deal fraction with guarantee*, *Reserve fund*, and *Overcollateralization* are given in %. Under the null hypothesis of the Wilcoxon rank-sum test the deals of the top 10% (5%) clients are distributed like the deals of the 90% (95%) smallest clients. Columns 3, 6, and 9 provide the standardized test statistics, which are approximately normally distributed in large samples. The symbols **, and *** represent *p*-values of below 10%, 5%, and 1%, respectively.

	Moody's			S&P			Fitch		
	Obs. (1)	Value (2)	z –Stat (3)	Obs. (4)	Value (5)	z –Stat (6)	Obs. (7)	Value (8)	z –Stat (9)
<i>AAA subordination</i>									
All deals	3199	33.12		3655	34.99		1843	24.17	
Deals of top 10% clients	1842	26.06	11.85***	2168	28.17	12.52***	992	19.89	6.54***
Deals of top 5% clients	1460	24.54	11.64***	1800	26.87	13.03***	676	16.77	8.23***
<i>Deal rating</i>									
All deals	3199	18.57		3655	21.23		1843	16.26	
Deals of top 10% clients	1842	13.68	12.26***	2168	15.49	12.63***	992	14.04	5.52***
Deals of top 5% clients	1460	13.69	10.85***	1800	14.49	13.52***	676	11.57	7.07***
<i>Delinquency rate</i>									
All deals	622	1.24		435	1.60		351	1.58	
Deals of top 10% clients	429	1.44	3.44***	318	1.90	3.27***	226	1.92	2.72***
Deals of top 5% clients	324	1.55	1.93*	247	2.19	3.74***	138	2.03	1.82*
<i>Deal fraction with guarantee</i>									
All deals	3199	5.97		3655	5.27		1843	4.90	
Deals of top 10% clients	1842	4.35	3.67***	2168	3.65	4.78***	992	3.37	3.52***
Deals of top 5% clients	1460	4.10	3.59***	1800	3.13	6.17***	676	2.75	3.65***
<i>Reserve fund</i>									
All deals	622	2.65		435	2.07		351	1.94	
Deals of top 10% clients	429	2.61	0.42	318	2.08	0.38	204	1.99	0.74
Deals of top 5% clients	324	2.68	0.37	247	2.12	0.26	124	2.02	0.47
<i>Overcollateralization</i>									
All deals	622	−0.26		435	−0.07		351	0.02	
Deals of top 10% clients	429	−0.21	2.02**	318	0.08	0.30	204	0.26	0.05
Deals of top 5% clients	324	−0.06	1.02	247	0.37	1.01	124	0.79	1.61
<i>No. of tranches</i>									
All deals	3199	4.70		3655	4.46		1843	4.78	
Deals of top 10% clients	1842	4.76	3.54***	2168	4.55	2.62***	992	5.16	1.95*
Deals of top 5% clients	1460	4.80	2.81***	1800	4.43	5.81***	676	5.34	0.52
<i>Deal face value</i>									
All deals	3199	1.30bn		3655	1.11bn		1843	1.56bn	
Deals of top 10% clients	1842	1.71bn	16.39***	2168	1.44bn	16.38***	992	2.14bn	12.24***
Deals of top 5% clients	1460	1.79bn	12.80***	1800	1.45bn	11.68***	676	2.52bn	12.01***

(b) Table 8: nonparametric test for rating favors.

6 Short summary

- The paper tests whether issuer-pay credit rating agencies grant rating favors to important structured-debt clients.
- The key relationship variable is bilateral shared securitization business between an issuer and a CRA.
- The paper constructs a deal-level rating statistic using rating-implied spreads so that tranche ratings can be aggregated in a cardinal way.
- The final regression sample contains 672 deals and 1,404 deal-CRA observations.
- Larger shared business is associated with better ratings after controlling for credit risk, deal characteristics, asset type, collateral nationality, time, and issuer fixed effects.
- The preferred coefficient implies that a one-standard-deviation increase in log shared business lowers the deal rating by about 41%.
- Rating favors are stronger for deals with more tranches, consistent with the idea that complex products make conflicts harder to detect.
- Rating favors are stronger during the 2004–2006 credit boom and weaker during the crisis period.
- All three major CRAs show evidence of relationship-based rating favors.
- The effect is present across asset classes and is not driven by a single CRA or asset category.
- Nonparametric tests show that top clients receive lower AAA subordination and better deal ratings, even though their collateral quality is not better.
- The broad conclusion is that issuer-CRA business relationships distorted structured-debt ratings during the securitization boom.

7 Conclusion

7.1 Core conclusion

Efing and Hau show that structured-debt ratings are systematically more favorable when issuers have stronger bilateral business relationships with the rating agency. The result survives extensive credit-risk controls and issuer fixed effects, making simple issuer quality an unlikely explanation.

7.2 Economic intuition

The intuition is the issuer-pay conflict of interest. Rating agencies depend on repeat business from structured-debt issuers. If a large issuer can move future business to a competing agency, the CRA has an incentive to cater to that issuer. The incentive is strongest when the product is complex and when the market is booming, because rating favors are harder to detect and reputational costs appear lower.

7.3 Takeaway

The paper is an important empirical contribution to the study of ratings inflation. It shows that conflicts of interest were not just anecdotal; they can be measured in deal-level ratings and linked to bilateral business relationships between issuers and CRAs. The key lesson is that information intermediaries can distort information production when their revenue comes from the party being certified.